



Specializing Foundation Models via Mixture of Low-Rank Experts for Comprehensive Head CT Analysis

Youngjin Yoo¹, Han Liu¹, Bogdan Georgescu¹, Yanbo Zhang¹, Sasa Grbic¹, Michael Baumgartner¹, Thomas J. Re¹, Jyotipriya Das¹, Poikavila Ullaskrishnan¹, Eva Eibenberger², Andrei Chekkoury², Uttam K. Bodanapally⁴, Savvas Nicolaou⁵, Pina C. Sanelli⁶, Thomas J. Schroepel⁷, Yvonne W. Lui³, and Eli Gibson¹

¹ Digital Technology and Innovation, Siemens Healthineers, Princeton, NJ USA; ² Department of Computed Tomography, Siemens Healthineers, Forchheim, Germany; ³ Department of Radiology, New York University, New York, NY USA; ⁴ University of Maryland Medical Center, Baltimore, MD USA; ⁵ Vancouver General Hospital, Vancouver, BC Canada; ⁶ Northwell Health, New York, NY USA; ⁷ UCHealth Memorial Hospital, Colorado Springs, CO USA

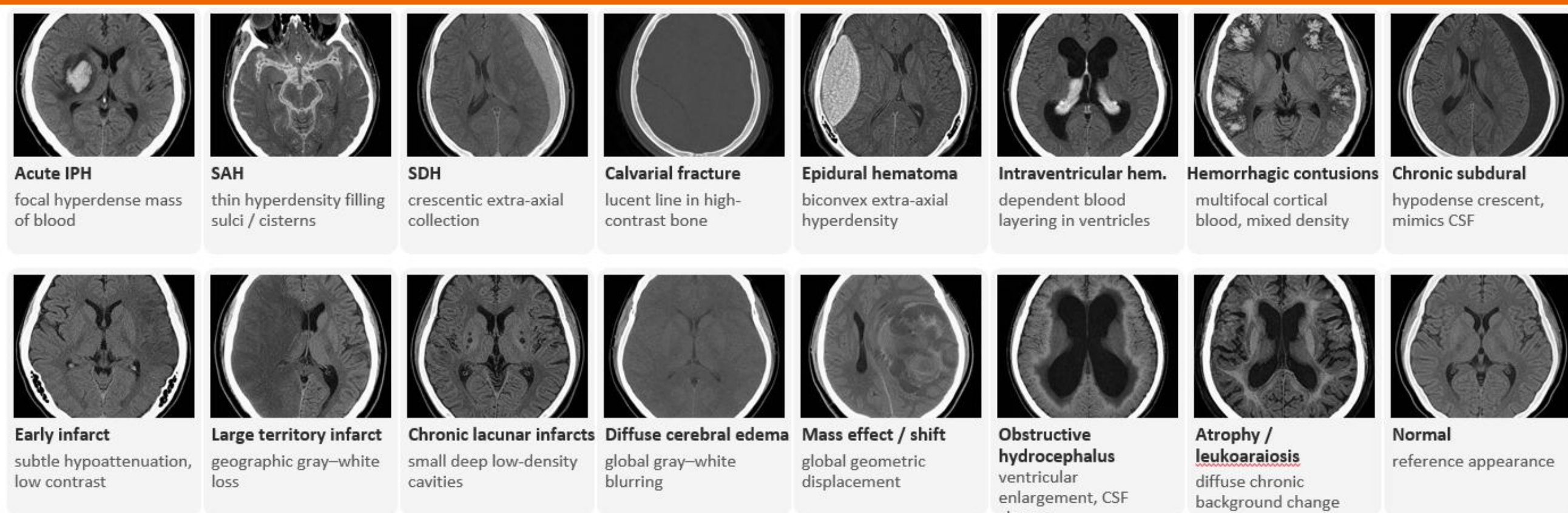
Motivation: Can a single adaptation serve all findings?

Foundation models transfer well on standardized benchmarks, but their behaviour in complex multi-label clinical scenarios remains insufficiently understood.

Standard LoRA applies a **uniform low-rank update across all inputs**, so acute hemorrhage, chronic ischemia, trauma and subtle structural findings compete for the same adaptation capacity — **knowledge interference**.

We propose **MoLRE**: multiple low-rank experts with unsupervised soft routing, enabling conditional adaptation with **< 0.5% additional parameters** and no explicit pathology supervision.

Head CT findings are visually heterogeneous

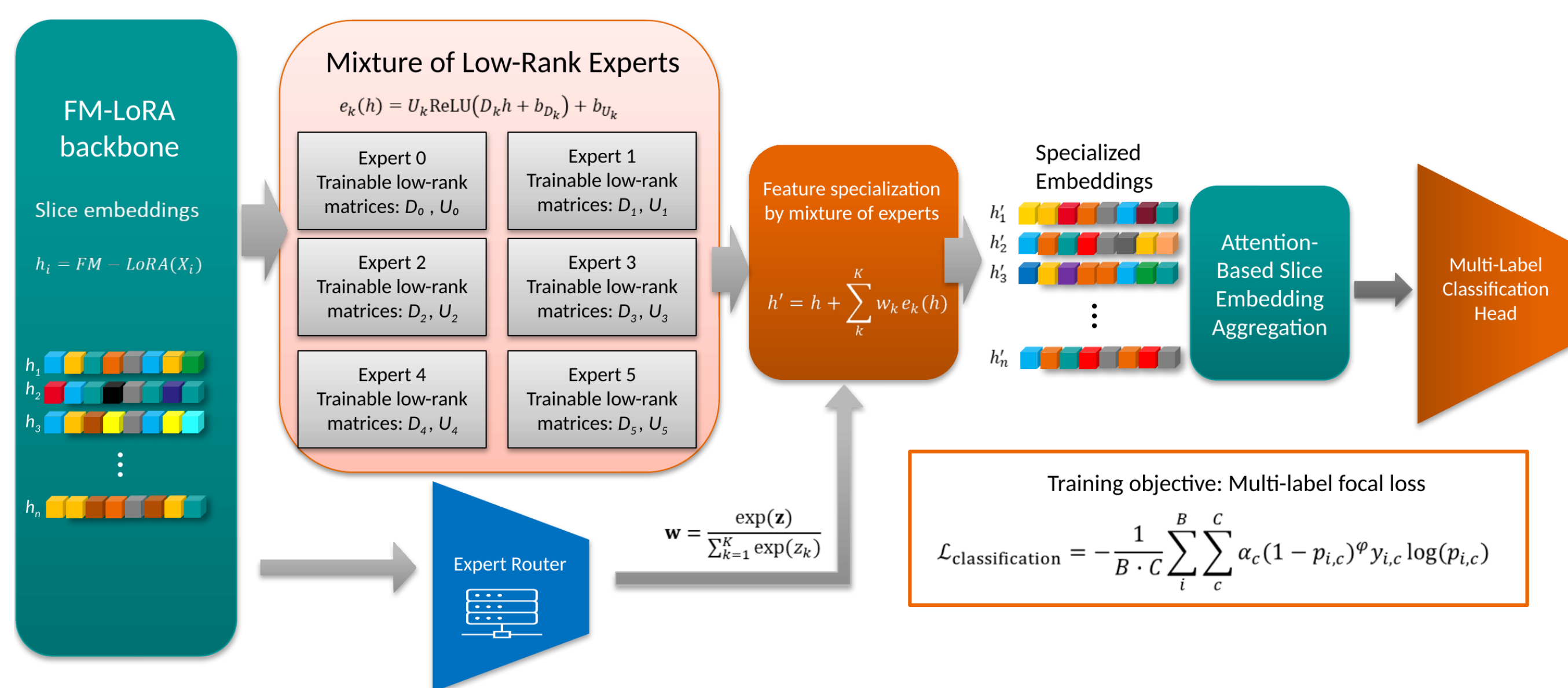


APPEARANCE hyperdense ↔ subtle hypoattenuation
COMPARTMENT parenchymal ↔ extra-axial ↔ bone
SCALE focal ↔ diffuse / global
DISPLAY brain ↔ subdural ↔ bone ↔ stroke

Findings span these axes simultaneously — a single low-rank update must encode all of them at once.

Illustrative synthetic images — not patient data, not for diagnostic use.

MoLRE: conditional low-rank specialization



$$h = W_0 x + \sum_i g_i(x) \Delta W_i x, \quad \Delta W_i = B_i A_i, \quad r \ll \min(d, d_{\text{out}})$$
$$g(x) = \text{Softmax}(W_2 \text{ReLU}(W_1 x + b_1) + b_2)$$

W_0 frozen · $K = 6$ experts, rank 8 · soft routing learned unsupervised from the task loss · slice-level in 2D, volume-level in 3D

Data and label construction

72,756 NCCT studies
9 centers, 3 vendors
74 findings, 6 categories



Labels	Private GPT-4-mini · batches of 15 findings · impression section only
Validation	618 annotated cases — 0.982 accuracy · 1.10 FP and 0.16 FN per case
Splits	patient-level: 58,251 train · 7,291 optimization · 7,214 held out

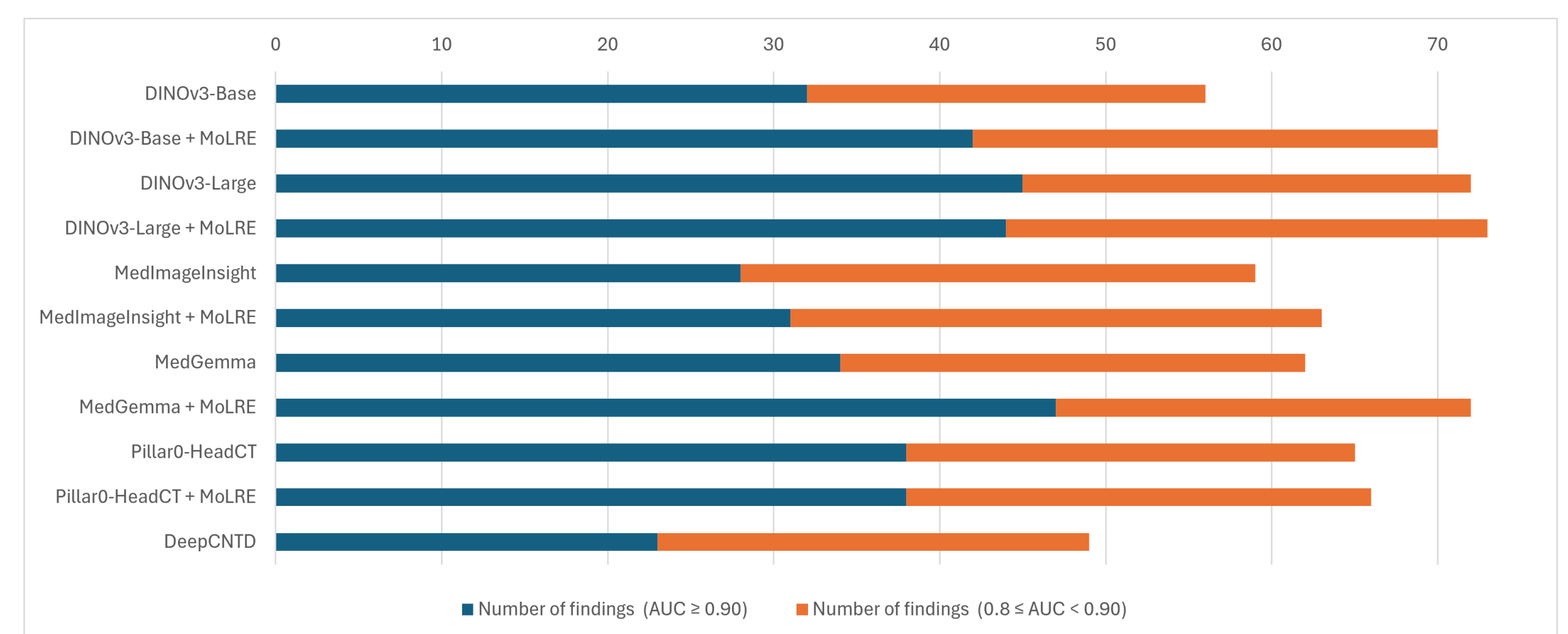
Benchmark with six foundation models

Model	Domain	Params	Base AUC	+ MoLRE	Δ	MoLRE params
DINOV3-Base	Natural	86.6M	0.855	0.902	+4.7%	0.28M
DINOV3-Large	Natural	305.2M	0.910	0.912	+0.2%	0.37M
MedImageInsight	Medical	364.3M	0.863	0.875	+1.2%	0.37M
MedGemma	Medical	431.1M	0.873	0.917	+4.4%	0.41M
Pillar0-HeadCT (3D)	Head CT	113.9M	0.891	0.893	+0.2%	0.41M
DeepCNTD	Head CT	6.9M	0.831	—	—	—

Mean AUC across 74 findings on 7,214 held-out studies. All 2D models use LoRA with attention-based slice aggregation; Pillar0-HeadCT is fully fine-tuned; DeepCNTD lacks attention pooling and is excluded from MoLRE.

MoLRE improves every compatible model, +0.2% to +4.7%; MedGemma + MoLRE reaches the best mean AUC of 0.917.

Stratified multi-finding detection performance



MoLRE mainly increases the number of findings in the high-confidence regime ($\text{AUC} \geq 0.90$) rather than uniformly shifting all classes: DINOV3-Base rises from 32 to 42 findings and MedGemma from 34 to 47.

Per-finding detection performance



DINOV3-Base vs + MoLRE

MedGemma vs + MoLRE

Largest gains occur for visually subtle, heterogeneous or underrepresented findings; findings with strong baselines such as major hemorrhage show limited headroom.

When does conditional specialization help?

DINOV3-Base	+4.7%
general 2D, natural	
MedGemma	+4.4%
medical VLM, generative	
MedImageInsight	+1.2%
medical VLM, discriminative	
DINOV3-Large	+0.2%
general 2D, higher capacity	
Pillar0-HeadCT	+0.2%
3D, head CT-specific	

Pretraining objective matters. Generative vision–language features (MedGemma) gain more than discriminative image–text features (MedImageInsight).

Where routing sits matters. In the 3D model routing acts after volumetric pooling, once slice-level heterogeneity is already compressed.

Targeted, not uniform. MoLRE lifts challenging or borderline findings rather than boosting all classes equally.

Robustness. Mean AUC is flat across $K \in \{3, 6, 9\}$ and rank $r_e \in \{4, 8, 12, 16\}$; inference adds < 100 ms and 0.58 GB per scan.

Limitations. Routing and added capacity are not yet disentangled; Effect of expert count, rank, and insertion depth; Expert specialization interpretation.

Conclusion. Adaptation benefit depends on the interaction between pretraining domain, architecture and model scale — not on model size alone.

Disclaimers:

The concepts and information presented in this paper are based on research results that are not commercially available. Future availability cannot be guaranteed.

Contact: youngjin.yoo@siemens-healthineers.com